A program for calculating convolutions using the input side algorithm is shown in Table 6-1. Remember, the programs in this book are meant to convey *algorithms* in the simplest form, even at the expense of good programming style. For instance, all of the input and output is handled in **mythical** subroutines (lines 160 and 280), meaning we do not define how these operations are conducted. Do not skip over these programs; they are a key part of the material and you need to understand them in detail.

The program convolves an 81 point input signal, held in array X[ ], with a 31 point impulse response, held in array H[ ], resulting in a 111 point output signal, held in array Y[ ]. These are the same lengths shown in Figs. 6-3 and 6-4. Notice that the names of these arrays use upper case letters. This is a violation of the naming conventions previously discussed, because upper case letters are reserved for frequency domain signals. Unfortunately, the simple BASIC used in this book does not allow lower case variable names. Also notice that line 240 uses a star for *multiplication*. Remember, a star in a program means multiplication, while a star in an equation means convolution. A star in text (such as documentation or program comments) can mean either.

The mythical subroutine in line 160 places the input signal into X[ ] and the impulse response into H[ ]. Lines 180-200 set all of the values in Y[ ] to zero. This is necessary because Y[ ] is used as an accumulator to sum the output components as they are calculated. Lines 220 to 260 are the heart of the program. The FOR statement in line 220 controls a loop that steps through each point in the input signal, X[ ]. For each sample in the input signal, an inner loop (lines 230-250) calculates a scaled and shifted version of the impulse response, and adds it to the array accumulating the output signal, Y[ ]. This nested loop structure (one loop within another loop) is a key characteristic of convolution programs; become familiar with it.

```
100  'CONVOLUTION USING THE INPUT SIDE ALGORITHM
110                                '
120 DIM X[80]                      'The input signal, 81 points
130 DIM H[30]                      'The impulse response, 31 points
140 DIM Y[110]                     'The output signal, 111 points
150                                '
160 GOSUB XXXX                     'Mythical subroutine to load X[ ] and H[ ]
170                                '
180 FOR I% = 0 TO 110              'Zero the output array
190   Y(I%) = 0
200 NEXT I%
210                                '
220 FOR I% = 0 TO 80               'Loop for each point in X[ ]
230   FOR J% = 0 TO 30           'Loop for each point in H[ ]
240     Y[I%+J%] = Y[I%+J%] + X[I%]*H[J%]
250   NEXT J%
260 NEXT I%                        '(remember, * is multiplication in programs!)
270                                '
280 GOSUB XXXX                     'Mythical subroutine to store Y[ ]
290                                '
300 END
```

TABLE 6-1